

Xander Rancap

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MASTER RESUME — comprehensive project and skills reference (not for direct submission)

PROFESSIONAL SUMMARY

Full-stack software developer and graduate of SAIT's Software Development program, based in Calgary. Experienced in designing, developing, and testing web applications across React, Next.js, Node.js, Python, and C#/.NET, with production experience in relational databases (PostgreSQL, MariaDB) and NoSQL databases (MongoDB). Brings a disciplined testing and quality assurance practice to every project, including automated end-to-end and unit test coverage (Cypress, Jest), load and performance testing (JMeter), continuous integration pipelines (GitHub Actions), and root-cause defect analysis. Has delivered AI-integrated features using the Anthropic API and Groq's Llama 3, and maintains production systems currently in daily use. Comfortable owning a project from initial design through deployment and ongoing support, and collaborating effectively within Agile teams.

EDUCATION

Southern Alberta Institute of Technology (SAIT)

Calgary, Canada

Diploma — Software Development, 3.61 GPA | 2024 – 2026

- Relevant coursework: Software Security, Object-Oriented Programming I–III (Python, C#, Java), Web Development (JavaScript, TypeScript, React), Database Systems (SQL), Cloud Computing (Azure).

PROJECTS

Resoniq — Guitar Tone Analysis Platform

2026 – Present

Next.js 16, React 19, TypeScript, Tailwind CSS, Framer Motion, Python, FastAPI, Librosa, PostgreSQL, Prisma, Supabase Storage, Docker Compose

- Designed and built a full-stack app end to end: a guitarist uploads a song and gets back an inferred tone recipe — amp voicing, cabinet, pickup position, EQ, gain, and effects chain — based on real audio-feature extraction rather than guesswork.
- Built the frontend in Next.js 16, React 19, and TypeScript, handling authentication, file upload, and an interactive, animated dashboard for tuning the result, and built a separate Python/FastAPI microservice using Librosa to do the actual signal processing.
- Documented the heuristics layer honestly as a starting point for a player to dial in by ear, not a verified gear match. Stored data in PostgreSQL via Prisma, files in Supabase Storage, and containerized the whole stack with Docker Compose.

F1Dash — AI-Powered F1 Analytics Platform

January 2026 – Present

Next.js, TypeScript, Groq (Llama 3), React Query, Recharts, OpenF1 API, Jest

- Designed and developed a full-stack sports analytics platform featuring live telemetry displays, driver comparisons, and interactive data visualizations, consuming third-party REST APIs with pagination and search functionality across a dataset of 350+ race records.
- Integrated Groq's Llama 3 language model via API to generate AI-driven race insights from live structured data, and developed a context-aware chatbot capable of answering user questions using real-time prediction data, including the underlying prompting logic and conversation flow.
- Implemented a weighted prediction engine incorporating recent form, qualifying pace, and circuit history, producing probability-based outputs, and authored Jest unit tests for the prediction calculations to identify computational errors prior to UI rendering.

YYC Track — Calgary CTrain Rating Platform | SAIT Capstone Project

September 2025 – April 2026

React, TypeScript, Node.js/Express, MongoDB, Azure AI, Cypress, Jest, GitHub Actions, Jira — Agile team environment

- Developed the React/TypeScript frontend for a multi-user civic feedback platform as part of an Agile development team, collaborating directly with UX and backend team members to implement designs and define API contracts.
- Authored automated regression test suites in Cypress covering core user workflows, along with Jest unit test coverage for critical application logic, executed on a per-commit basis through a GitHub Actions continuous integration pipeline.
- Performed root-cause analysis on reported defects, documented reproduction steps, expected versus actual results, and severity in Jira, then validated fixes and conducted re-testing prior to ticket closure. Communicated test status and blockers through daily stand-ups and sprint planning.
- Integrated Azure AI Content Safety and sentiment analysis services to automate moderation of user-submitted comments. Delivered the application fully deployed on Microsoft Azure and presented the finished product at SAIT CapCon 2026.

WMBA — Where My Bus At?

February 2026 – Present

Next.js 16, Express, TypeScript, Prisma, PostgreSQL, Leaflet, Recharts, GTFS Real-time

- Built an experimental full-stack project to sharpen DevOps and backend skills, ingesting Calgary Transit's live GTFS protobuf feed every 60 seconds to track 600+ active buses across 530+ routes.

- Built the Express/TypeScript backend with Prisma and PostgreSQL to store real-time vehicle positions, calculate speeds using Haversine distance, and maintain a rolling 25-hour vehicle history table powering trend analytics.
- Built a Next.js 16 frontend with a dark, command-center-style interface — a Leaflet map with heatmap overlays, 1-hour vehicle trails, and route shape polylines, alongside an analytics dashboard with speed distributions, route comparisons, and hourly trend charts in Recharts, all sourced from Open Calgary's public GTFS endpoints.

BMR Pharmacy Sales Tracker & Website

November 2025 – Present

React, Vite, TypeScript, Node.js, Express, PostgreSQL (Supabase), Tailwind CSS, Cypress, Jest, Ollama (local LLM)

- Replaced a pharmacy's manual, paper-based sales records with a full-stack system providing live revenue dashboards, ranked product summaries, and automated monthly reporting; the system remains in daily production use.
- Authored Cypress end-to-end tests and Jest unit tests covering core dashboard workflows, and validated computed revenue and product aggregates directly against source transaction records in PostgreSQL rather than relying solely on rendered UI output.
- Own production defect triage for the system since launch, including reproducing user-reported issues, isolating root cause, and verifying fixes prior to release. Conducted cross-browser regression testing and evaluated the consistency of a locally hosted LLM-powered assistant (Ollama) across representative input sets.

ApexF1 — Python ML Race Prediction Platform

January – April 2026

Python (Machine Learning), Next.js, React, Express, PostgreSQL (Supabase)

- Developed the complete web application layer around a Python-trained machine learning prediction model, including PostgreSQL database schema design, REST API routes, and a React interface presenting win and podium probability outputs alongside historical model accuracy.
- Tested REST API responses under incomplete, delayed, and malformed input conditions, confirming that the system degraded predictably rather than failing silently.

FitZone Gym Management

2024

.NET MAUI Blazor Hybrid, C#, MariaDB

- Led a small development team in building a full-stack gym management application in C# using .NET MAUI Blazor Hybrid, covering secure authentication, membership tracking, and scheduling functionality, from initial setup through to a working, demoable product.
- Designed and implemented the MariaDB relational database and its integration with the application, and coordinated task planning and delivery across the team.

Cloud Security, Compliance & Performance Testing | Azure Coursework

2025

Docker, Terraform, Azure Container Apps, API Management, Key Vault, JMeter

- Executed load and performance testing using JMeter against a containerized REST API, validating rate-limiting thresholds and system behavior under sustained concurrent request volume.
- Verified security controls, including IP filtering, CORS policy, HTTPS enforcement, and customer-managed encryption keys, confirming correct behavior under both valid and invalid request conditions.
- Validated containerized deployments provisioned through Terraform Infrastructure-as-Code, confirming environment parity across rebuilds.

EXPERIENCE

First Eduspec Inc.

Makati, Metro Manila, Philippines

Immersion (Intern) | 2024

- Programmed more than 15 Arduino circuits using ROBOTC and collaborated on a greenhouse automation prototype through the full development cycle, presenting the completed solution to an evaluation panel and earning an innovation award.

SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C#, Java, SQL

Frontend Development: React, Next.js, Vite, .NET MAUI Blazor Hybrid, Tailwind CSS, HTML5/CSS3, Recharts, Leaflet, React Query, Context API

Backend & Data: Node.js, Express, Python (FastAPI), REST API design, Prisma ORM, PostgreSQL, MongoDB, MariaDB, Supabase, Supabase Storage

AI & LLM Integration: Anthropic API, Groq (Llama 3), Ollama (local LLM), prompt engineering, AI-driven feature design, audio-feature extraction (Librosa)

Test Automation: Cypress (end-to-end testing), Jest (unit testing), JMeter (load and performance testing), GitHub Actions CI execution

Test Methodology: Functional, integration, regression, and performance testing; white-box and black-box test design; root-cause defect analysis; defect documentation; requirements review

Cloud & Infrastructure: Microsoft Azure, Azure AI, Docker, Terraform, Vercel, Render

Tools & Process: Git, GitHub, Jira, Figma, Agile/Scrum methodology, sprint planning, code review, technical documentation